VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS305

Theory of Computation and Automata - Laboratory Manual & Practical Guide

Syllabus Focus: CS305 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Finite Automata & Regular Languages

" Syllabus Scope: DFA, NFA, epsilon-NFA, subset construction algorithm, minimization of DFA Myhill-Nerode

Unit 2: Regular Language Properties & Pumping Lemma

" Syllabus Scope: Pumping lemma for regular languages, closure properties union, intersection, complement

CS305: Theory of Computation and Automata - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Context-Free Grammars & Pushdown Automata

" Derivation trees, ambiguity, Chomsky Normal Form CNF, Greibach Normal Form GNF, PDA transition mechanics

Unit 4: Context-Free Language Limits & Pumping Lemma

" Pumping lemma for CFLs, Ogden lemma, closure properties of CFLs, CYK parsing algorithm

Unit 5: Turing Machines & Undecidability

" Standard Turing Machine model, multi-tape TMs, Church-Turing thesis, Universal Turing Machine, Halting probl

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Regular Expression Engine and Lexical Tokenizer Simulator in Python

Lab 2: DFA Simulator and State Minimization Algorithm Implementation

Lab 3: Pushdown Automaton Simulator for Palindrome and Balanced Bracket Recognition

Lab 4: Single-Tape Turing Machine Simulator for Binary Addition and Multiplicative Unary

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John E. Hopcroft, Rajeev Motwani - Introduction to Automata Theory, Languages, and Computation Pearson

* Michael Sipser - Introduction to the Theory of Computation Cengage Learning 3rd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students